their objectives, in their utter lack of intent to do harm, and in the brilliance of their design and performance, these robots speak eloquently for us.

But being much more advanced scientists and engineers than we—otherwise they would never be able to find and retrieve the small, silent spacecraft in interstellar space—perhaps the aliens would have no difficulty understanding what is encoded on these golden records. Perhaps they would recognize the tentativeness of our society, the mismatch between our technology and our wisdom. Have we destroyed ourselves since launching *Voyager*, they might wonder, or have we gone on to greater things?

Or perhaps the records will never be intercepted. Perhaps no one in five billion years will ever come upon them. Five billion years is a long time. In five billion years, all humans will have become extinct or evolved into other beings, none of our artifacts will have survived on Earth, the continents will have become unrecognizably altered or destroyed, and the evolution of the Sun will have burned the Earth to a crisp or reduced it to a whirl of atoms.

Far from home, untouched by these remote events, the *Voyagers*, bearing the memories of a world that is no more, will fly on.

---

*It takes so long to circuit the Sun because its orbit is so vast, 23 billion miles around, and because the force of the Sun's gravity—which keeps it from flying out into interstellar space—is at that distance comparatively feeble, less than a thousandth what it is in the Earth's vicinity.

*Robert Goddard, the inventor of the modern liquid-fueled rocket, envisioned a time when expeditions to the stars would be outfitted on and launched from Triton. This was in a 1927 afterthought to a 1918 handwritten manuscript called "The Last Migration." Considered much too daring for publication, it was deposited in a friend's safe. The cover page bears a warning: "The[se] notes should be read thoroughly only by an optimist."

*The Earth, by definition, is 1 AU from its star, the Sun.

*Radio signals that both *Voyagers* detected in 1992 are thought to arise from the collision of powerful gusts of solar wind with the thin gas that lies between the stars. From the immense power of the signal (over 10 trillion watts), the distance to the heliopause can be